

晶圆上 μ LED 和微型显示器的快速测试

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为测量应用提供定制解决方案

显示屏 /
智能设备测量



AR/VR



汽车内部照明



显示特性

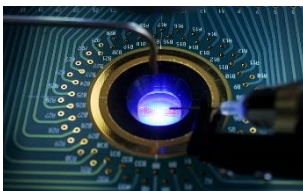


平板显示器生产

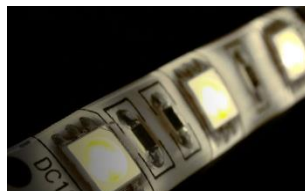


系统审计

光发射器
测量



LED 生产测量



LED / 模组



UV-LEDs



VCSEL / Laser

照明产品
测量



汽车外部照明



蓝光危害



固态照明测量

Premium German Quality in Light & Display Measurement.

MicroLED
Quantum Dots
Curved Display
VCSEL
OLED UV-LED
AR/VR Glasses
Blue Light Hazard
Flat Panel Display
Smart Devices
Laser IR
Automotive Interior
Vehicle Headlights
SSL Luminaires
Head Up Displays
Retroreflectors

We bring Quality to Light

Global Sales Network and Service Teams.



Agenda

1. 简介

2. 晶圆光致发光测试

3. μ (O)LED 显示屏电致发光测试

1. 单LED/像素分析

5. 峰值波长与主导波长的对比

6. 执行时间分析

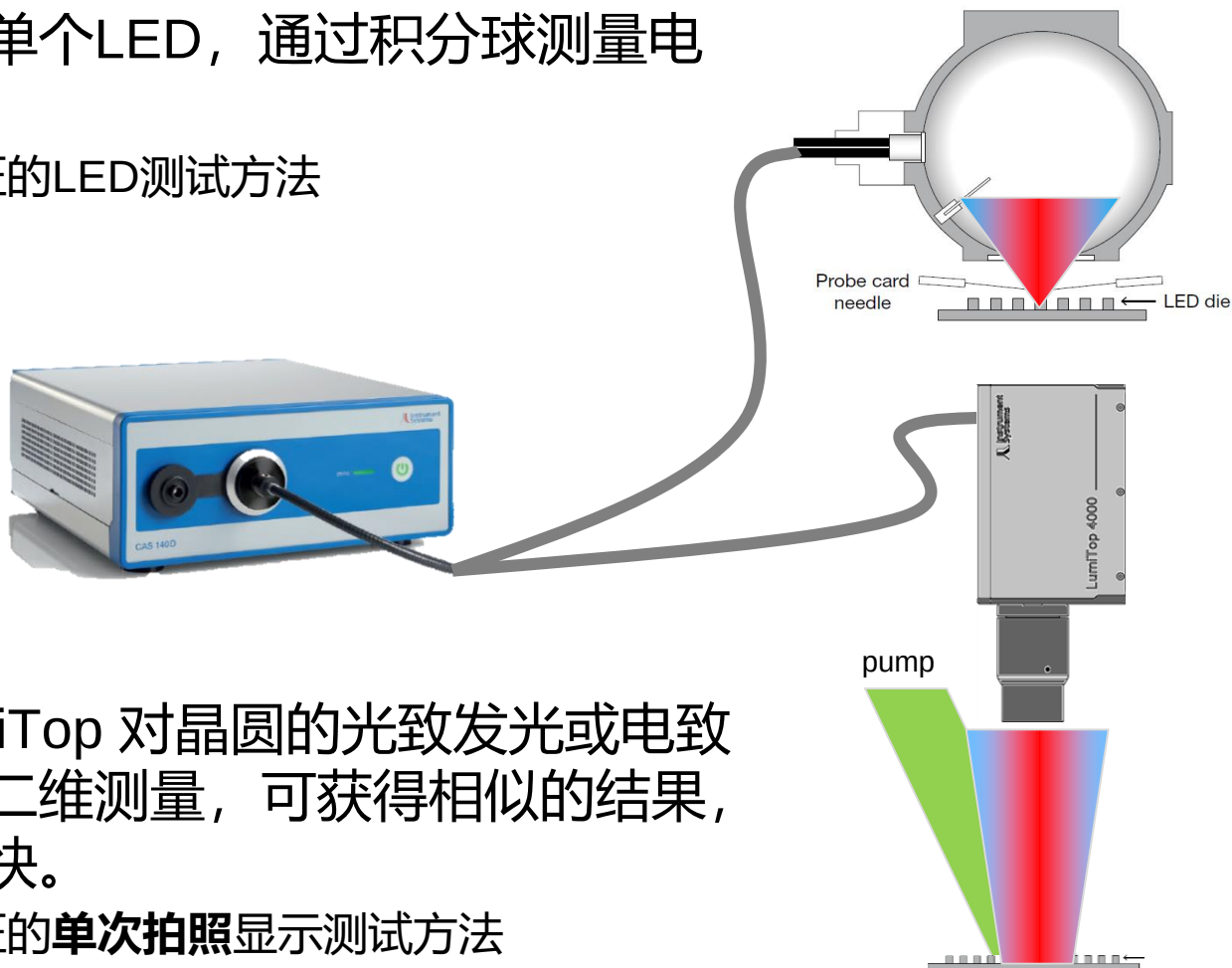
7. 总结

μLED 晶圆测试

▲ 逐颗点亮单个LED，通过积分球测量电致发光

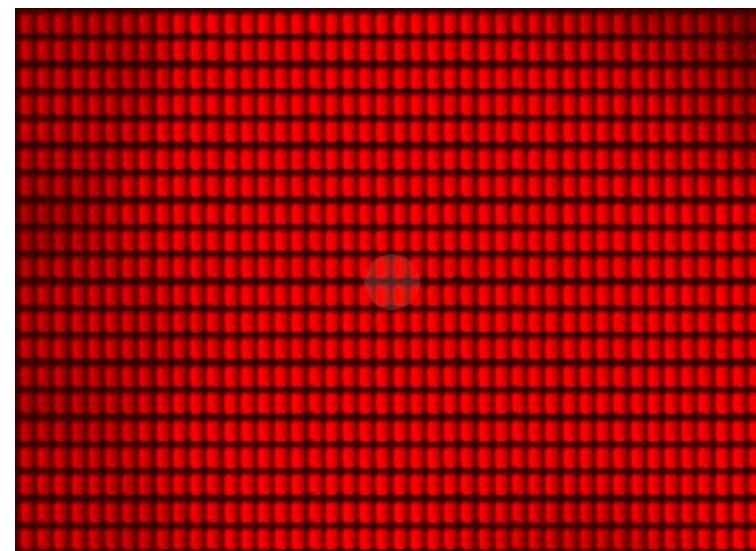
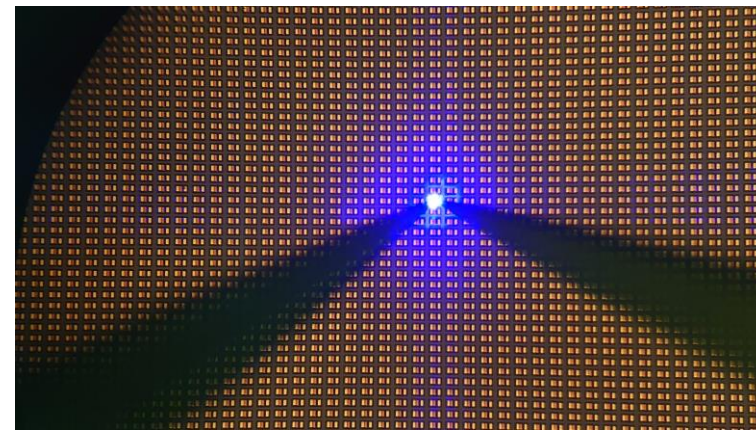
- 经验证的LED测试方法

光谱仪

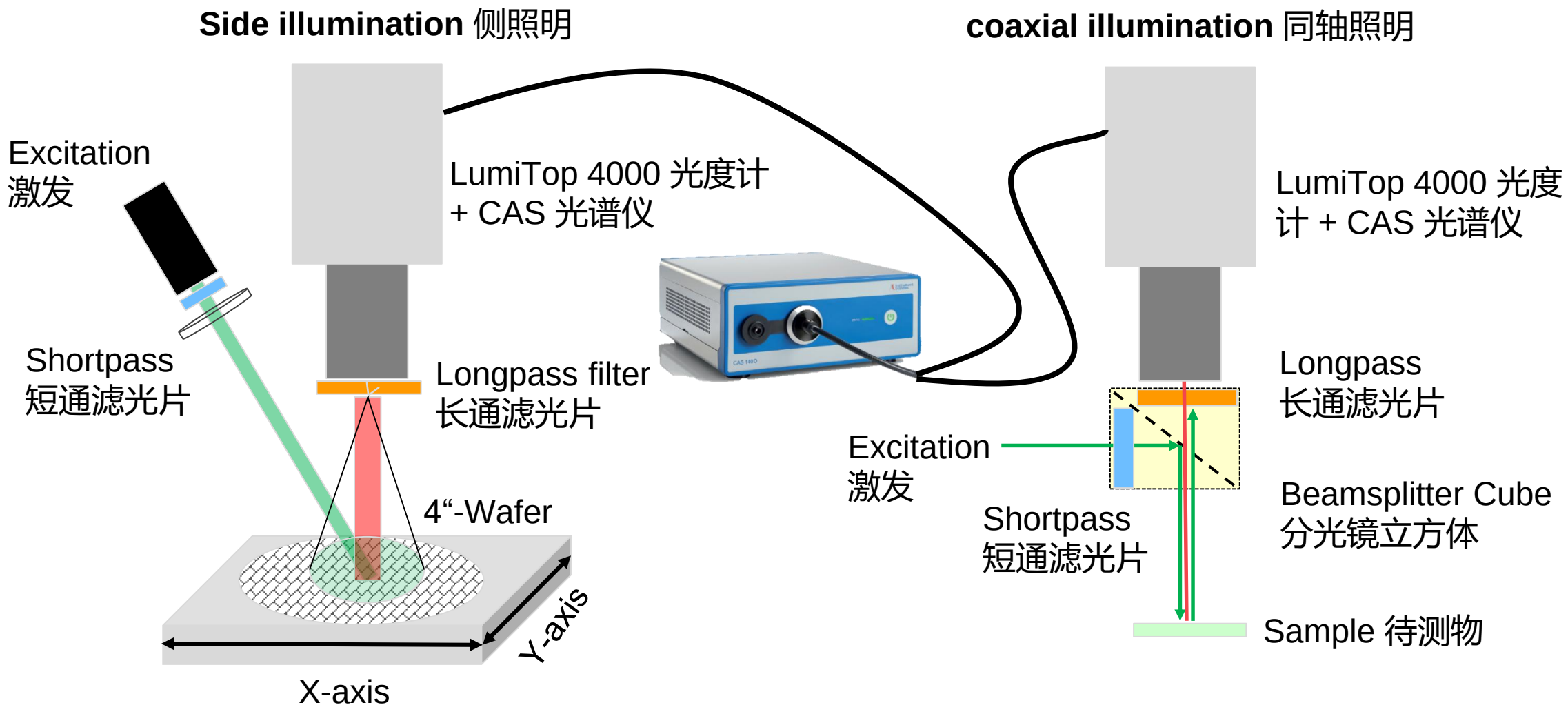


▲ 使用 LumiTop 对晶圆的光致发光或电致发光进行二维测量，可获得相似的结果，但速度更快。

- 经验证的**单次拍照**显示测试方法



光致发光测量典型配置

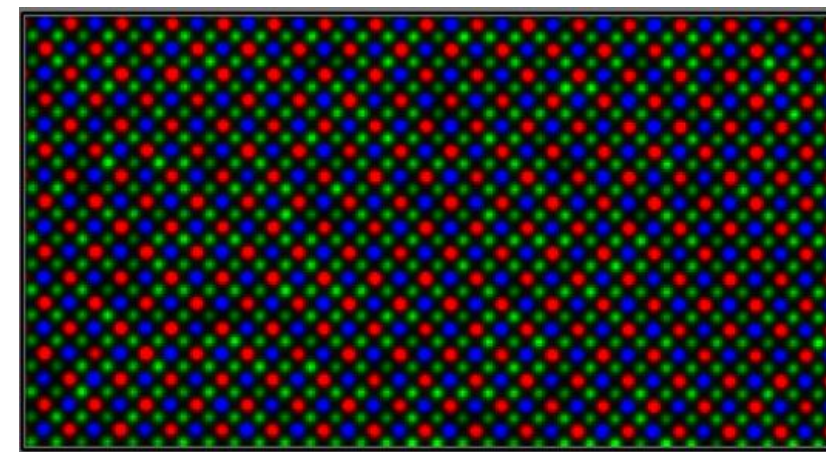
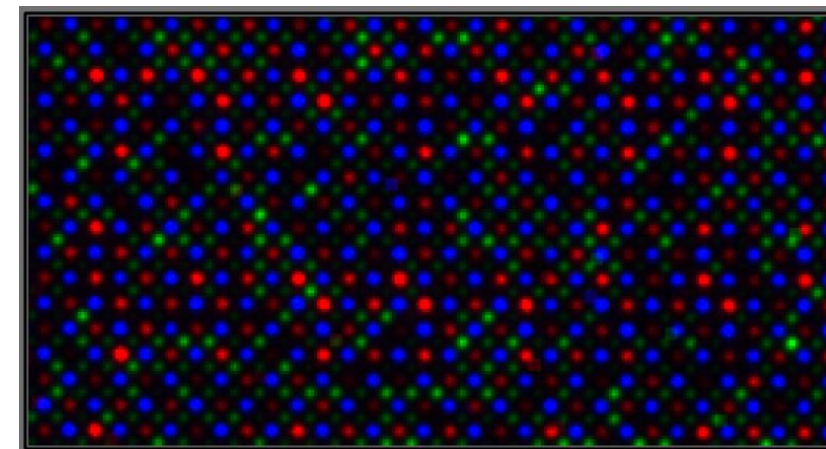
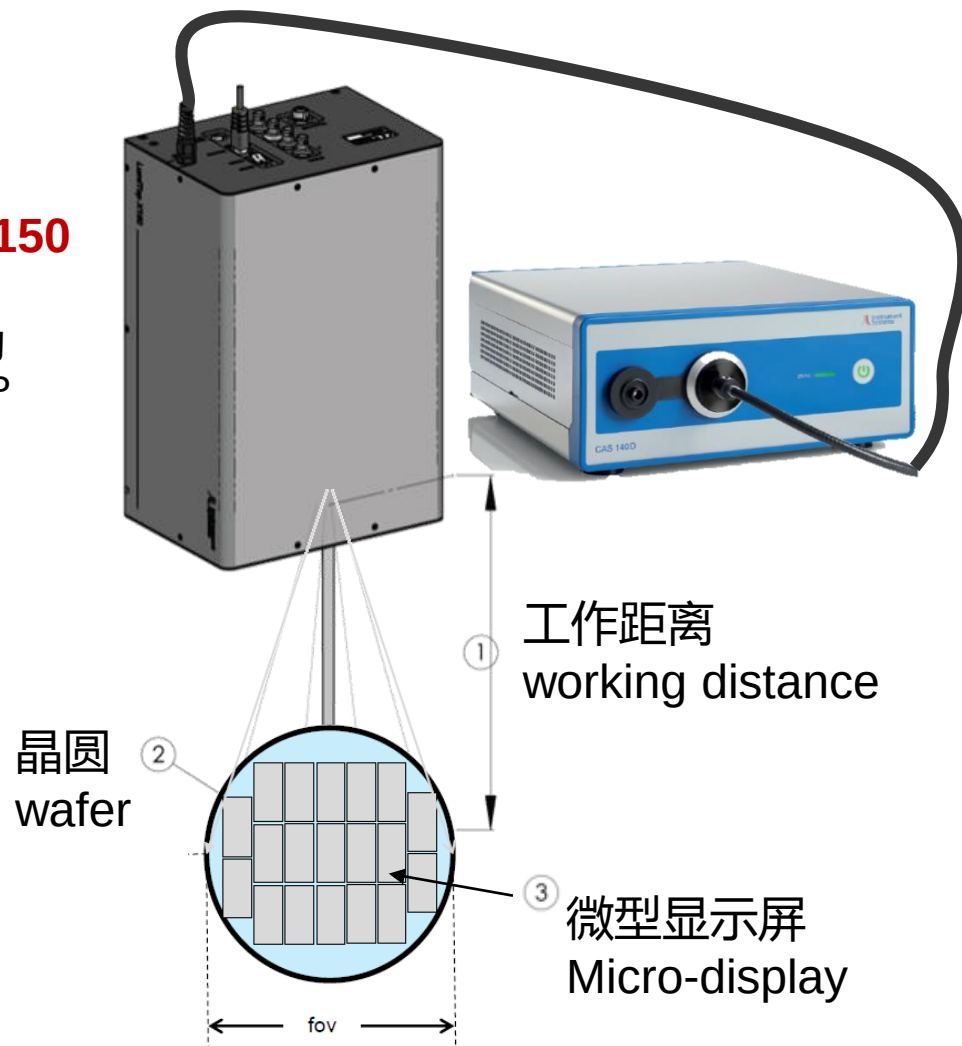


晶圆上的 μ (O)LED显示测试典型设置

LumiTop X150

600MP

单个颜色通道的
分辨率为150MP



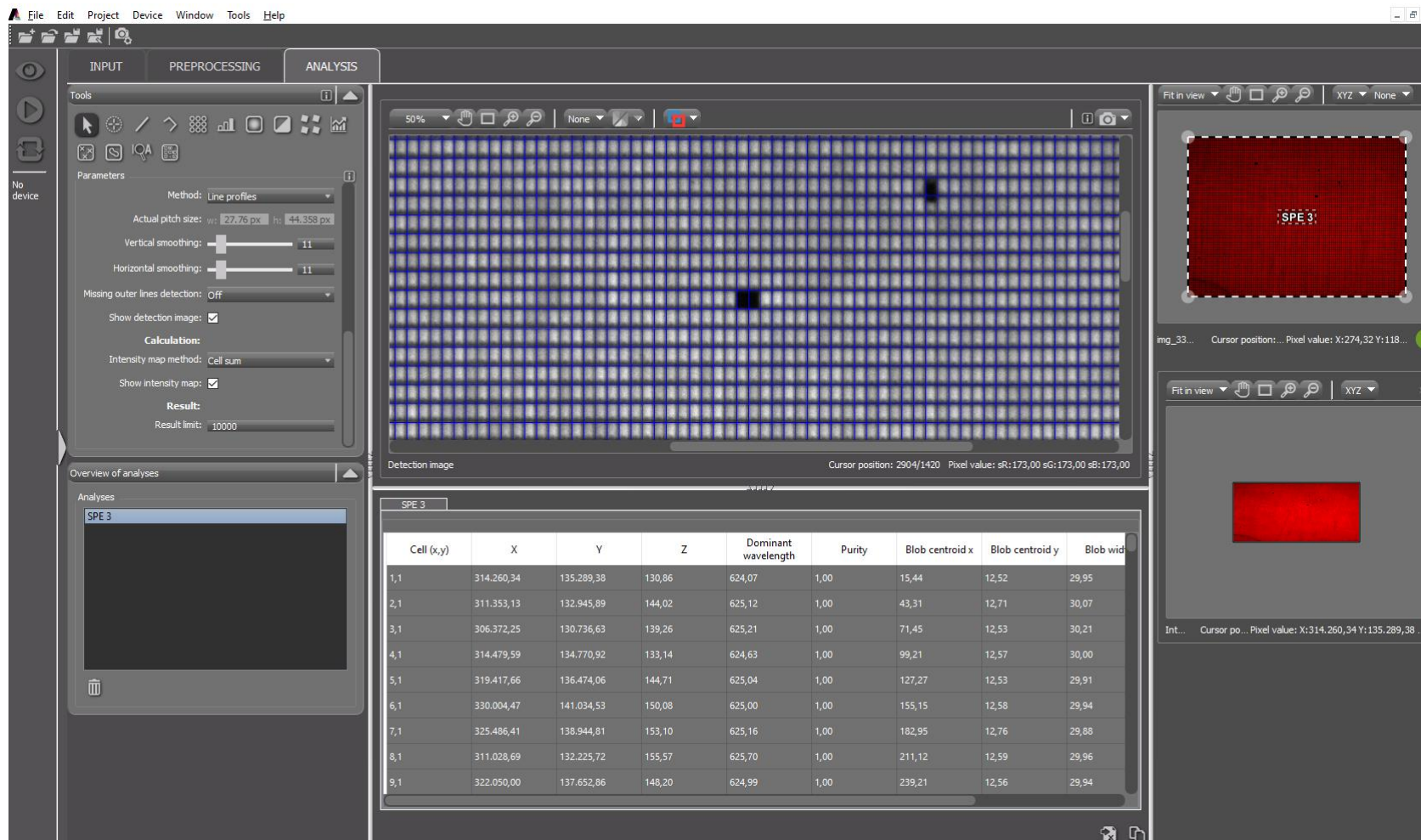
LumiSuite 单像素评估 (Single Pixel Evaluation)

SPE 分析:

检测到包含单发射极的单元格网格

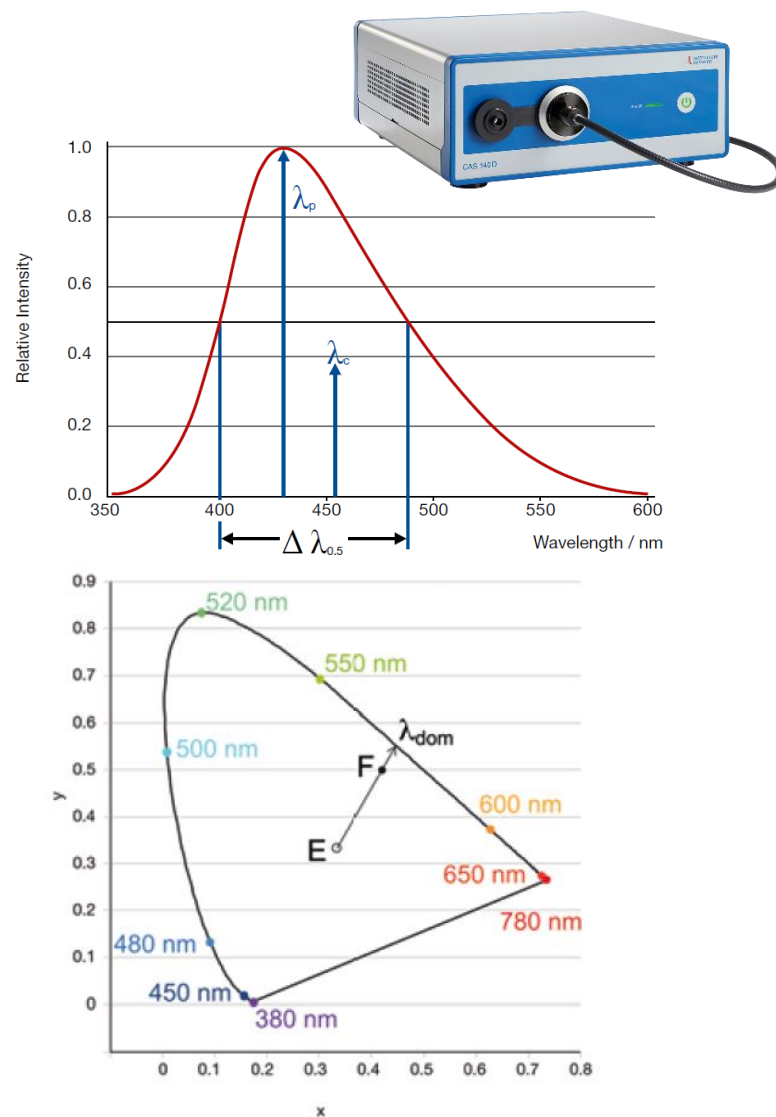
SPE 结果参数:

- 单元格位置
- 三刺激值: $X, Y, Z \rightarrow Lxy$
- 主波长 Dominant wavelength
- 纯度 Purity
- 发射斑点的位置和面积:
 - 中心点 x, y
 - 高度
 - 宽度



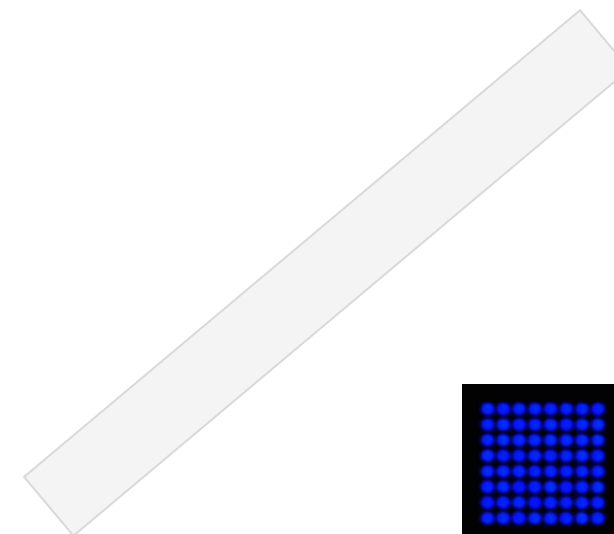
峰值波长 vs. 主波长

- 显示计量标准：主波长和色度是评估二维应用中色彩感知的重要参数
- 照明计量标准：峰值波长是一维 LED 辐射测量应用中的重要参数
- LumiTop 可以非常精确地测量主波长的二维分布。峰值波长可在光谱仪的测量区域内获得。
- 研究表明，峰值波长和主波长与LumiTop的设置之间存在很强的相关性。

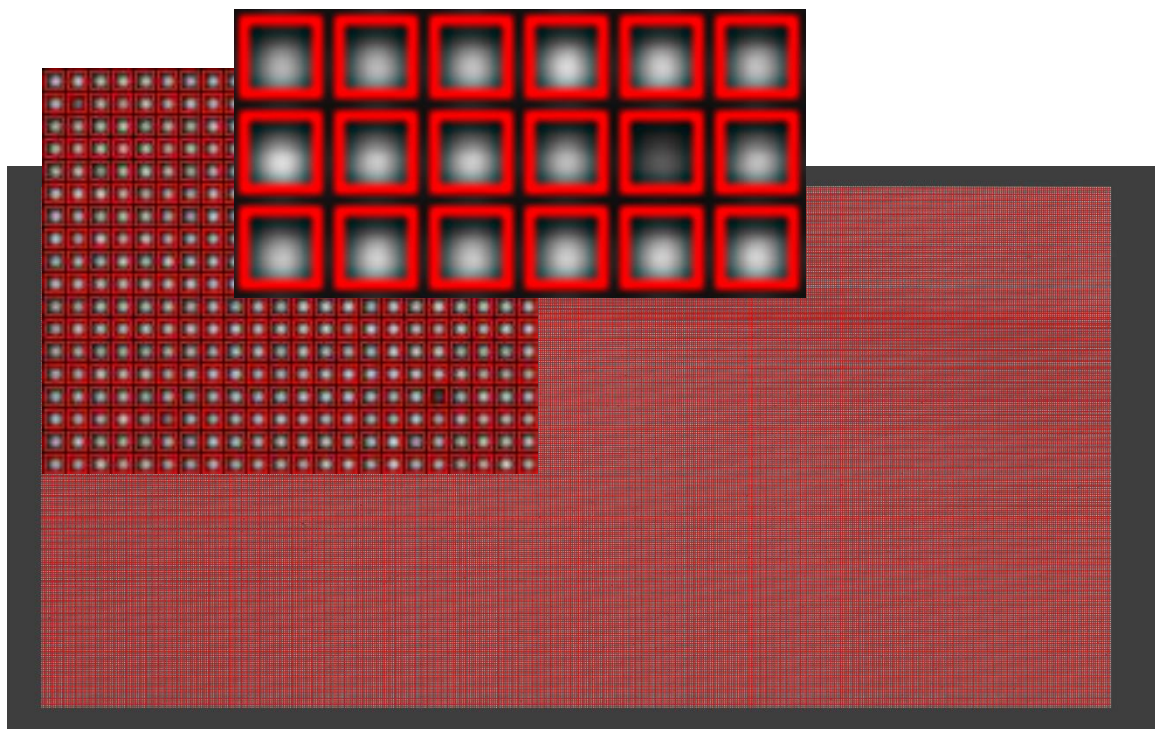


光谱仪的峰值波长 λ_{peak}

- ▲ 数据显示峰值波长与主波长之间存在相关性
- ▲ 通过添加相应的平均偏移量（对于单色光，偏移量为零），可以很好地从主波长 λ_{DOM} 估算出每个 LED 的峰值波长 λ_{PEAK}
- ▲ 相关性在 $\pm 0.3 \text{ nm}$ 纳米范围内

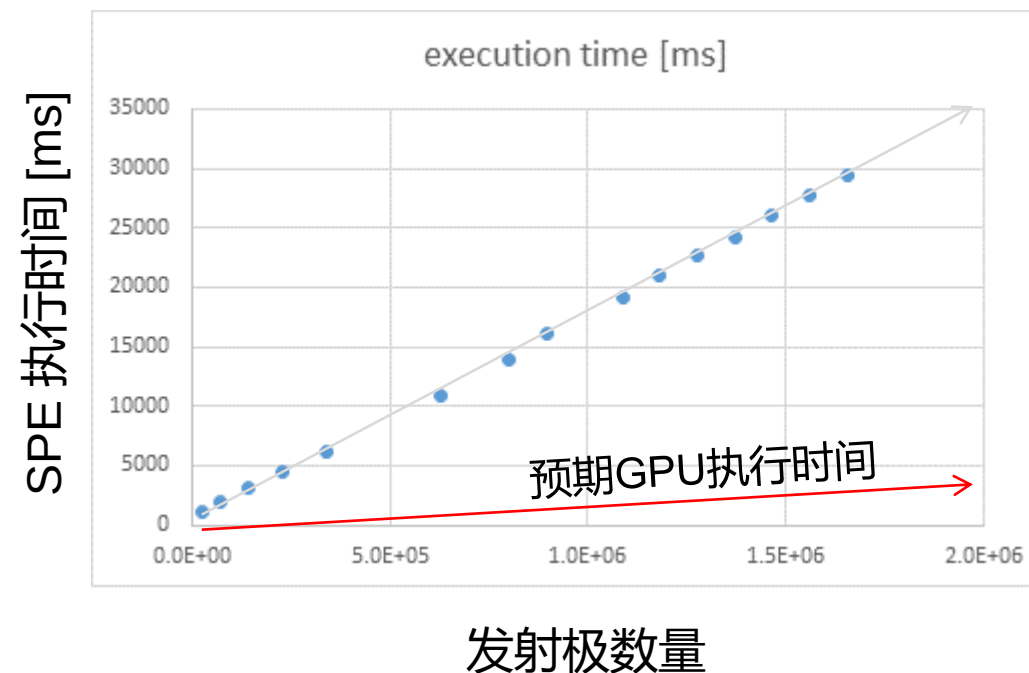


单像素评估速度: μ OLED 显示屏



$1.7 \cdot 10^6$ 发射极, 发射器尺寸 $5\mu\text{m}$ / 间距 $11\mu\text{m}$

CPU 执行时间: $12\text{-}25\mu\text{s}$ / LED



测量速度模拟器

测量速度的计算

模拟 4 英寸晶片 LumiTop 的测量速度

可按具体情况模拟和优化测量速度性能

关联于 LED 规格 (间距、尺寸、亮度)

Device (LT4000 + 100mm lens)		
Device size in px	Horizontal	4096
	Vertical	3000
FOV size in um	Horizontal	14400
	Vertical	10500
Resolution um/px	Horizontal	3,5
	Vertical	3,5
CAS Spot Diam. um		1000
CAS Spot Diam. Px		284,4
Working distance		257
Tact times(ms)	Setup	20
	Hw trigger setup	57
	Download	162
	Overhead	5
Minimum ROI	Horizontal	128
	Vertical	2
Minimum ROI increment	Horizontal	16
	Vertical	2
Readout & dload speed	Horizontal (ms/px)	0,000011
	Vertical (ms/row)	0,01

DUT		
Number of uLEDs per measure	columns	67
	rows	45
Size of uLEDs in um	Horizontal	34
	Vertical	58
Size of uLEDs in px	Horizontal	9,7
	Vertical	16,6
Size of gap in um	Horizontal	25
	Vertical	30
Size of gap in px	Horizontal	7,1
	Vertical	8,6

Probe		
Maximum simultaneous uLED	Horizontal	67
	Vertical	45
Checker pattern separation	Horizontal	1
	Vertical	1

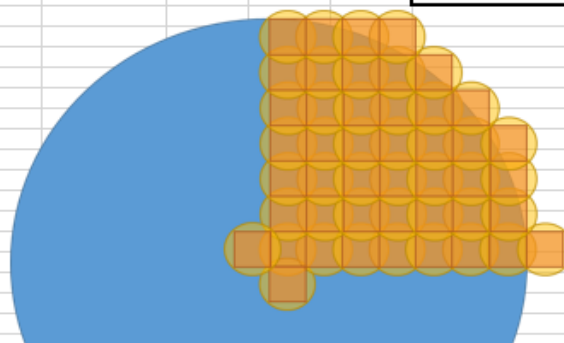
Wafer		
Wafer Radius	inches	2
	mm	50,8
	px	14450
Illuminated Spot radius	mm	2,8
	px	796
Square size (inside circle)	px	1126
Number of squares		553

FOV		
FOV (%)	Horizontal	27%
	Vertical	38%
Usable FOV (%)	Horizontal	70%
	Vertical	100%
Meas. FOV (px)	Horizontal	1126,3
	Vertical	1126,3
CAS FOV fill (%)		48%
Minimum CAS FOV (%)		50%
CAS FOV fill (%)	Horizontal	393%
	Vertical	395%
How many uLED inside cas spot (Rough calc)		86

Algorithm (SPE)		
Minimum gap px		3
Time per uLED m		0,018

Takt time estimation (ms)				
Use hardware trigger?	No			
CAS measurement everytime	Yes			
Setup	20			
Wait for trigger	1			
Camera Integration	6			
Download	25,2			
Overhead	5			
Calibration	22,7			
Algorithm (SPE)	54,27			
Handler tact time	0			
CAS Integration	19			
Nr. of measurements	553			
Single measurement (ms)	134			
Full DUT	74	7	0,04	-
Scenario 1	74	7	0,04	0%
Scenario 2	63	9	0,04	15%
Scenario 3	63	9	0,04	15%
Scenario 4	63	9	0,04	15%

Warnings				
FOV orientation incorrect	0	<-- horizontal orientation is faster than		
CAS Fill factor insufficient				
Handler speed is bottleneck				
Algorithm is bottleneck				
CAS integration is bottleneck				
DUT too big for FOV	0			
Insufficient gap resolution	0	<-- change checker pattern to avoid		



二维 μ LED 测量应用

- ▲ μ LED 汽车外部照明
 - LumiTop 4000 +100 mm 镜头
- ▲ μ /mini-LED 汽车内部照明
 - LumiTop 4000 + 广角镜头
- ▲ μ (I/O)LED Displays 显示屏
 - LumiTop X150 (600MP)
- ▲ μ LED 晶圆光致发光测试
 - LumiTop 测量系统: 12MP - 600MP

05 \\ LED wafer and array testing

LED-based light sources have largely superseded conventional light sources. Due to their technical advantages, they have given rise to many new exciting applications. In particular, upcoming μ LED technologies are expected to drive displays in head-mounted devices for augmented and virtual reality applications and other mobile devices. Each LED in the millions of pixels on a wafer is a single light source with individual variations in color and luminance. To assure the highest quality under the constraints



of economical production takt times, fast and accurate optical testing has to be provided for every LED on the wafer posing new challenges to LED and μ LED production testing.

LumiTop 4000, 32mm and 130 mm, optionally with periscope.

2D testing of μ LED wafers

With millions and millions of LEDs on a single wafer, standard sequential testing of each μ LED with an integrating sphere is not practical anymore. A massively parallelized 2D testing process for defects and spectral properties is the key to an economical production. The LumiTop concept has been extended to innovative wafer testing by using a dedicated macro lens and a specifically optimized calibration

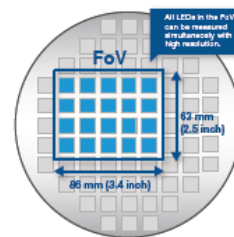
for narrow bandwidth μ LEDs. A hardware trigger for synchronization ensures reproducible measurements of time dependent light sources and speeds up measurement times.

- ▲ Fast 2D testing of μ LED wafers and arrays
- ▲ Defect detection
- ▲ Optimized color calibration
- ▲ High luminance options
- ▲ Trigger for synchronization

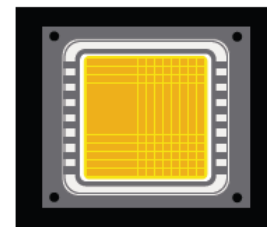
Automotive μ LED array testing: Adaptive headlights

Adaptive headlights in cars are becoming bright, high resolution LED-projectors. For safety reasons and quality control these light sources need to be tested rigorously to meet automotive specifications.

LumiTop provides the accuracy and customization to provide testing solutions also for very bright μ LED arrays.



Wafer Testing: FoV customizable according to μ LED size.



Schematic μ LED array for adaptive automotive front lighting systems.

μLED 计量方法总结

▲ 光电或电致发光的图像

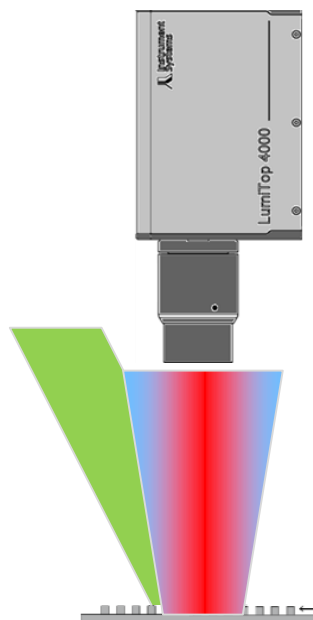
- 单像素/μLED 检测
- 参考光谱 (可溯源至 NIST/PBT)

▲ 单像素评估 (人眼感知)

- 色度 □ 主波长
- 峰值波长可以由 λ_{dom} 推导出
- 亮度
- 纯度
- CIE三刺激值函数: X,Y,Z

▲ 空间评估

- 均匀性
- LED / 像素位置
- 单元格大小和发光面积



▲ CPU 执行时间:

- 12-25μs / LED

▲ GPU 预期执行时间

- <1 μs /LED
- 2k 显示屏 : 2s
- 4“晶圆 (4×10^7 μLED): 40秒

感谢您的关注!

联系我们:

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